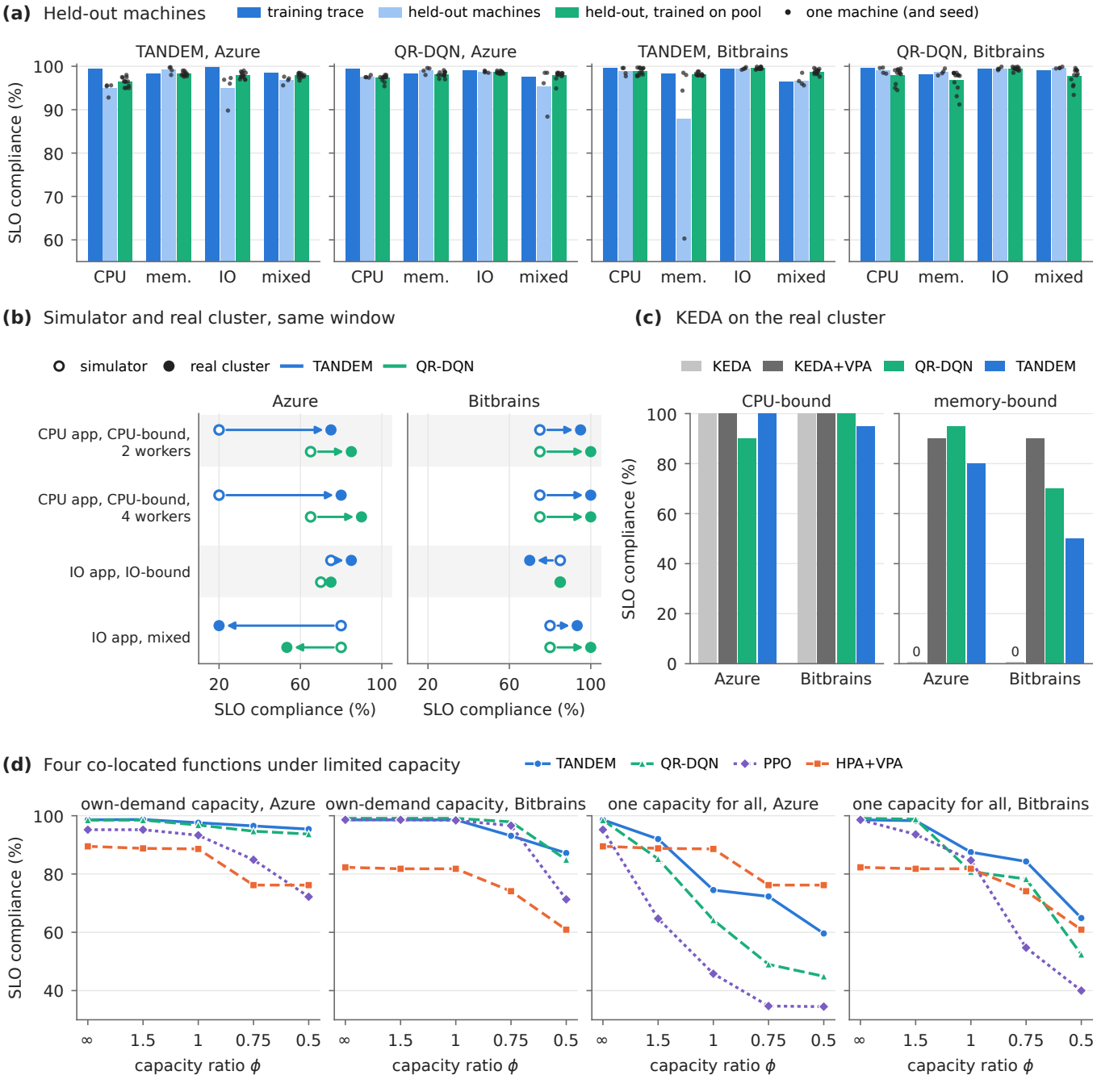
(a) Held-out machines
training trace
held-out machines
held-out, trained on pool
one machine (and seed)
TANDEM, Azure
QR-DQN, Azure
TANDEM, Bitbrains
QR-DQN, Bitbrains
SLO compliance (%)
CPU
mem.
IO
mixed
(b) Simulator and real cluster, same window
simulator
real cluster
TANDEM
QR-DQN
Azure
Bitbrains
CPU app, CPU-bound, 2 workers
CPU app, CPU-bound, 4 workers
IO app, IO-bound
IO app, mixed
SLO compliance (%)
(c) KEDA on the real cluster
KEDA
KEDA+VPA
QR-DQN
TANDEM
CPU-bound
memory-bound
(d) Four co-located functions under limited capacity
PPO
HPA+VPA
own-demand capacity, Azure
own-demand capacity, Bitbrains
one capacity for all, Azure
one capacity for all, Bitbrains
capacity ratio ϕ